

Charlie Loitman, EIT

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EDUCATION

- University of California, Berkeley**, Master's of Climate Solutions May 2026
- Interdisciplinary degree focusing on power systems, data science, grid management, and electrification
 - Co-lead BERC Symposium on water markets and BERC Summit on demand response technologies
 - Relevant coursework: Data Science for Energy, Energy and Environmental Markets, Electric Transmission Systems
- Yale University** – *cum laude* | *Tau Beta Pi* | *GRE: Verbal 168, Quant 170, Writing 6.0* December 2022
- ABET-Accredited B.S. in Mechanical Engineering *with distinction*; B.A. in Economics; Certificate in Energy Studies
 - Engineering Capstone: A Method to Reduce Carbon Emissions from Concrete Usage in 3D Printing
 - Economics Capstone: Hotelling vs Reality: Understanding the Dynamics of Global Oil Production

EXPERIENCE

- Graduate Intern** | **Carbon Containment Lab**, New Haven, CT January 2026 – Present
- Analyzing New Mexico oil well data using scikit-learn and PyTorch to identify abandoned oil wells and predict instances of methane leakage to prioritize plugging. Developing an economic royalty payment model to incentivize marginal wells to shut down.
- Independent Study** | **SunStrong Management**, Berkeley, California January 2026 – Present
- Applying ML for >\$1B asset manager who owns residential solar assets at >500k homes to identify most profitable homes to install batteries. Using proprietary and ERCOT data from GridStatus to identify locations with greatest effects on reliability and congestion.
- Graduate Student Instructor** | **UC Berkeley**, Berkeley, California July 2025 – Present
- Write assignments and section materials for “Decision Analysis, Modeling, and Quantitative Methods” and “Marine Geochemistry.”
- Trader** | **Flow Traders**, Manhattan, New York February 2023 – March 2025
- Automated and standardized futures-based ETFs; focused on commodities and the expansion into carbon credits, cryptocurrencies, and active energy strategies. Systematization of trading strategies and research reduced operational burden from six traders to two, doubled trading volumes, and increased profit 40% y-o-y. Transitioned to co-create Centralized Execution Desk, developing strategies and a macro signal framework to improve trading across all products and optimizing order sizing and risk for each product and exchange. Became youngest member of global Trader Tech Team, coordinating product development and technology initiatives.
- Summer Analyst** | **Goldman Sachs, Customized Beta Strategies**, Manhattan, New York May – August 2022
- Analyzed performance of risk models to allow tracking of custom benchmarks (S&P500, etc.) while tailoring investment goals (eg decarbonization). Created simulation tool used distributed to hundreds of financial advisers to demonstrate results across scenarios.
- Research Analyst** | **Ellington Management Group**, Stamford, CT January – August 2021
- Conducted mathematical research to find probability density functions of credit default swaptions to develop new options strategy. Presented research in PhD seminar weekly and developed a VIX trading strategy through options arbitrage.
- Undergraduate Learning Assistant** | **Yale University**, New Haven, CT August 2019 – December 2023
- Held office hours, problem-solving sessions, and reviews for Proof-Based Linear Algebra, Proof-Based Vector Calculus, Thermodynamics, Fluid Dynamics, Dynamics, and Aerodynamics. I taught lectures for Aerodynamics while professor travelled.
- Undergraduate Researcher** | **MCubed Lab**, WashU, MO and **Qiu Lab**, Yale, CT May 2019 – December 2020
- Applied ML classification with MCubed to identify novel stable high entropy alloys and double perovskites as candidates for solar panel materials. Won **George J. Schulz Fellowship in Physical Science** to fund DFT analysis of promising crystals for Qiu Lab.

AWARDS

- 4th Place** | **UMich Ross Banking Design Dilemma**, Theme: Banking strategies for grid expansion February 2026
- 2nd Place Winner** | **Rice Business Energy Case Competition**, Theme: Datacenter power demand November 2025
- L.C. Lichty and E.O. Waters Prize in Mechanical Engineering** | **Yale University** May 2023
- “To the graduating senior who most demonstrates high scholarship and original research in mechanical engineering.”
- Award for Outstanding Undergraduate Achievement** | **Yale Science & Engineering Association** April 2023

Languages: Computer: Python, C++, MATLAB, SQL, Julia. Modelling: PyPSA, PyTorch, Gurobi. Spoken: Spanish (intermediate)

Affiliations: Berkeley Energy and Resources Collaborative, Haas Climate Solutions Fund, Brooklyn Greenway Initiative